

Jamie Moritz Holdcroft

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EDUCATION

University of New South Wales (UNSW)

Bachelor of Science (Mathematics Major, Physics Minor) | Bachelor of Computer Science

Expected Aug 2027

Sydney, Australia

- Weighted Average Mark (WAM): **82.87** Distinction
- **Selected Coursework:** Real Analysis, Complex Analysis, Discrete Mathematics, Multivariable Calculus, Linear Algebra, Quantum Physics, Classical Mechanics, Electromagnetism
- **Selected Advanced Coursework (Postgraduate-level):** Extended Algorithm Design & Analysis (HD), Computer Vision (HD), Machine Learning & Data Mining (DN)

Technical University of Munich (TUM)

Visiting Student (Exchange)

Apr-Sep 2026

Munich, Germany

- Awarded DAAD STIBET Scholarship, funded by the German Academic Exchange Service
- **Selected Advanced Coursework (Postgraduate-level):** Mathematical Modeling and Simulation, Measure-theoretic Probability Theory and Statistics

Reddam House Sydney, Australia

Higher School Certificate (ATAR: 95.9)

2018-2023

Sydney, Australia

- **Leadership and Awards:** Australian Defence Force (STEM) Future Innovators Award (2023), Liebman Sustainability Award (2023), Head of Environmental Portfolio, Elected to School Executive Representative Council

PUBLICATIONS & RESEARCH OUTPUT

Abdallah, A., **Holdcroft, J.**, Ali, M., & Jatowt, A. (2026). *Are LLM-Based Retrievers Worth Their Cost? An Empirical Study of Efficiency, Robustness, and Reasoning Overhead. Proceedings of the 49th International ACM SIGIR Conference (SIGIR 2026 Reproducibility Track).*

Abdallah, A., **Holdcroft, J.**, Jatowt, A., *et al.* (2026). *Rankify: A Comprehensive Python Toolkit for Retrieval, Re-Ranking, and Retrieval-Augmented Generation. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026 System Demonstrations Track)*

Holdcroft, J., Abdallah, A., & Jatowt, A. (2026). *The Magnitude Mirage: Rethinking Confidence for Reasoning-Intensive Retrieval.* Submitted to *ACL Rolling Review (ARR)*. **Under Review**

Holdcroft, J., Abdallah, A., & Jatowt, A. (2026). *Difficulty-Gated Fusion of Reasoning Views for Temporal Retrieval.* Submitted to *CIKM 2026*. **Under Review**

Holdcroft, J., Dewynne, J., & Howison, S. (2026). *Self-Referential Forward-Start Options.* **In Preparation**

EXPERIENCE

Research Intern (Information Retrieval & Machine Learning)

Jan-Apr 2026

University of Innsbruck (UIBK), Digital Science Center, Austria

- Produced research published and submitted to multiple A* venues with PhD researchers and faculty.
- Developed and extended retrieval methods in *Rankify*, a modular IR toolkit.
- Evaluated systems with new and standard IR metrics, analysing trade-offs in quality, robustness, and efficiency.

Research Project: Self-referential Forward-Start Options

Dec 2025-Present

Research Collaboration with University of Oxford Faculty

- First author of a theoretical paper on pricing self-referential forward-start options with Oxford researchers.
- Derived and analysed the implicit strike equation, characterising admissible solutions.
- Proved unimodality and identified the unique arbitrage-stable strike via perturbation-based analysis.

Research Project: Quantum Approximate Optimisation Algorithm

Sept-Nov 2025

University of New South Wales (UNSW), Sydney

- Studied parameter concentration phenomena in the QAOA for the Max-Cut problem across graph classes.
- Built a scalable grid-search pipeline for systematic exploration and benchmarking of QAOA parameter landscapes.
- Group supervised by Dr. Gerald Huang, achieving a final grade of 100% (highest in cohort).

Casual Academic

Jan 2025-Present

UNSW Computer Science and Engineering

- Recommended by lecturer and head academic staff to lead weekly undergrad laboratory classes in C programming.

- Collaborated effectively with academic staff to achieve 100% satisfaction rate in the MyExperience survey.

Independent School Canteen Operator

Reddam House Sydney, Australia

Jan 2022–Aug 2023

- Co-founded and independently ran the school canteen while an active student catering to 350+ staff and students.
- Generated national media coverage, featuring on 9 News Australia, 2GB Radio, Radio NZ, the Sydney Morning Herald and more.

SELECTED PROJECTS

- **Time-Series Forecasting (ML):** Developed ensemble models (RF, Ridge, GB) with feature engineering (EW-MAs, lags, FFT, PCA) for multi-horizon prediction on environmental climate data.
- **Dead-Tree Segmentation from UAV NIR Imagery (CV):** Developed and compared rule-based, Random Forest on superpixels, and U-Net CNN approaches; achieved 67% IoU on aerial imagery.
- **Navigation System (C):** Implemented Dijkstra-based routing on road-maps with time-dependent constraints and dynamic road closure updates.
- **Huffman Compression (C):** Built an end-to-end lossless compression system using optimal prefix-free encoding.

TECHNICAL SKILLS

Languages & Tools: C, Python, Bash, Linux, Git, Jupyter, L^AT_EX, Maple

ML / AI Frameworks: PyTorch, HuggingFace, scikit-learn, OpenCV, NumPy

INTERESTS

Poetry, philosophy, environmental sciences, surfing, alpine hiking, jazz